

RL Prediction:

Reinforcement Learning - Lecture 6

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PUCRS

Material adapted from:
Martha and Adam White - University of Alberta/Coursera
Sutton and Barto - Reinforcement Learning: An Introduction (2nd ed.)
David Silver - Reinforcement Learning

Outline

- 1 Recap
- 2 Monte-Carlo Learning
- 3 Temporal-Difference Learning
- 4 Comparison

- Last lecture:
 - Bandit problems
 - Exploration and exploitation in single decisions
- This lecture (Sutton and Barto 5.1.5.2 and 6.1-6.3)
 - **Prediction**
 - Monte-Carlo and Temporal Difference Learning
- Next lecture
 - **Control**
 - Comparison with Monte Carlo methods

Recap

- We understand how to plan in stochastic environments for both short-term and long-term goals via:
 - MDP formalization
 - Dynamic Programming algorithms
- These formalisms assume we have both $\mathcal{R}$ and $\mathcal{P}$
- But how can we make decisions when we have neither of these?
 - In some problems do not know the environment transition probabilities $\mathcal{P}$
 - Even when we have them, computation can be error-prone and tedious

Prediction and Control

Two key tasks in Reinforcement Learning as we move forward:

- Prediction: compute or estimate value and q-value functions
a.k.a. passive reinforcement learning
- Control: decide which action to take, given (or not), value functions
a.k.a. active reinforcement learning

Prediction and Control

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$$v(s) \leftarrow \mathcal{R}_s + \left(\max_a \gamma \sum_{s'} \mathcal{P}_{ss'}^a * v(s') \right)$$

- Control: decide which action to take, given (or not), value functions
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- Control: decide which action to take, given (or not), value functions
a.k.a. active reinforcement learning

$$\pi(s) = \arg \max_a \sum_{s'} \mathcal{P}_{ss'}^a * v(s')$$

Outline

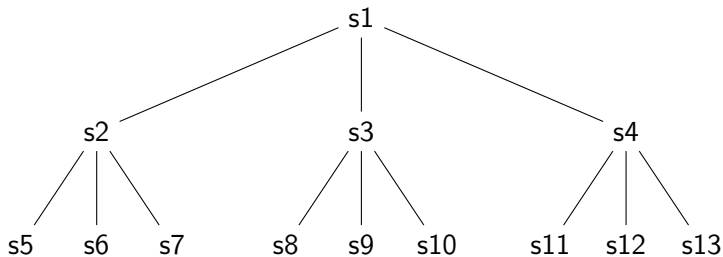
- 1 Recap
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Monte Carlo Reinforcement Learning

- MC methods learn directly from episodes of experience
- MC is model-free: no knowledge of MDP transitions / rewards
- MC learns from complete episodes: no bootstrapping
- MC uses the simplest possible idea: $\text{value} = \text{mean return}$
- Caveat: can only apply MC to episodic MDPs
 - All episodes must terminate

What is Monte Carlo?

- Monte Carlo methods are a class of algorithms for stochastic problems that:
 - rely on repeated random samples
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s1
|
s2
|
s6

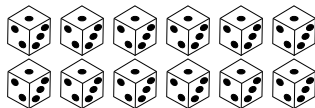
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 - rely on repeated random samples
 - in the limit approximate the sample mean
- Relies on a simulator rather than a complete probability model
- Each simulation is a **rollout**

s1
|
s3
|
s10

Monte Carlo Methods

What is the average value of 12-dice rolls?



Sum of Faces	Probability
12	?
13	?
14	?
...	
71	?
72	?

Monte Carlo Methods

Monte Carlo methods estimate values by **averaging** over a large number of random **samples**

What is the average value of 12-dice rolls?
Samples

$$\sum_{i=1}^{12} \text{die}_i$$


Sum of Faces	Probability
12	?
13	?
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...	
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72	?

39 38 49 36 33 47 37 49 41 40 44 44
35 43 47 28 46 52 47 45 43 43 42 43
40 32 29 48 43 49 48 39 42 35 44 48
36 32 41 49 34 54 37 42 37 38 42 50
39 37 45 49 44 34 38 50 35 36 42 45

Monte Carlo Methods

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40 32 29 48 43 49 48 39 42 35 44 48
36 32 41 49 34 54 37 42 37 38 42 50
39 37 45 49 44 34 38 50 35 36 42 45

Average=41.57

Monte-Carlo Policy Evaluation

- Goal: learn v_π from episodes of experience under policy π

$$S_1, A_1, R_1, \dots, S_k \sim \pi$$

- Recall that the **return** is the total discounted reward:

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

- Recall that the value function is the total discounted reward:

$$v_\pi(s) \doteq \mathbb{E}_\pi [G_t \mid S_t = s]$$

- Monte-Carlo policy evaluation uses **empirical mean** return instead of expected return



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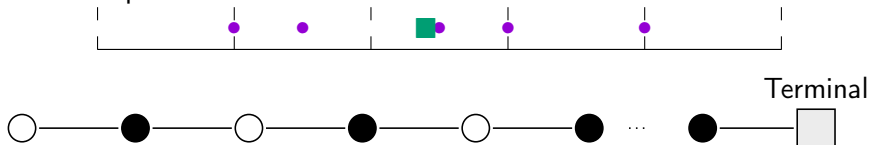
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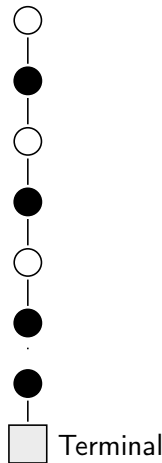
$$v_\pi(s) \doteq \mathbb{E}_\pi [G_t \mid S_t = s]$$

- Monte-Carlo policy evaluation uses **empirical mean** return instead of expected return



Backup diagram for Monte Carlo

- Entire rest of episode included
- Only one choice considered at each state (unlike DP)
 - thus, there will be an explore/exploit dilemma
- Does not bootstrap from successor states's values (unlike DP)
- Time required to estimate one state does not depend on the total number of states



First-Visit Monte-Carlo Policy Evaluation

- To evaluate state s
- The **first** time-step t that state s is visited in an episode,
- Increment counter $N(s) \leftarrow N(s) + 1$
- Increment total return $S(s) \leftarrow S(s) + G_t$
- Value is estimated by mean return $V(s) = S(s)/N(s)$
- By law of large numbers, $V(s) \rightarrow v_\pi(s)$ as $N(s) \rightarrow \infty$

Law of Large Numbers

When you take the mean of a bunch of i.i.d. random variables.

Law of Large Numbers

When you take the mean of a bunch of i.i.d. random variables.
This mean approaches the true expectation as you sample more.

Every-Visit Monte-Carlo Policy Evaluation

- To evaluate state s
- **Every** time-step t that state s is visited in an episode,
- Increment counter $N(s) \leftarrow N(s) + 1$
- Increment total return $S(s) \leftarrow S(s) + G_t$
- Value is estimated by mean return $V(s) = S(s)/N(s)$
- Again, $V(s) \rightarrow v_\pi(s)$ as $N(s) \rightarrow \infty$

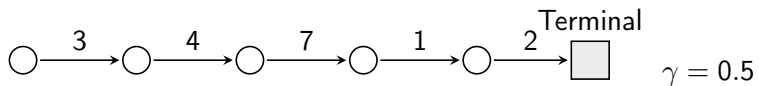
Monte Carlo Prediction Algorithms

Every-visit MC prediction

Require: a policy π to be evaluated

```
1: function mcPrediction( $\pi$ )
2:   Initialize  $V(s) \in \mathbb{R}$ , for all  $s \in \mathcal{S}$ 
3:    $Ret(s) \leftarrow$  an empty list, for all  $s \in \mathcal{S}$ 
4:   loop forever (for each episode)
5:     Generate an episode following  $\pi$ :  $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$ 
6:      $G \leftarrow 0$ 
7:     for each step of episode,  $t = T - 1, T - 2, \dots, 0$  do
8:        $G \leftarrow \gamma G + R_{t+1}$ 
9:       Append  $G$  to  $Ret(S_t)$ 
10:     $V(S_t) \leftarrow \text{average}(Ret(S_t))$ 
```

Computing returns efficiently



$$G_0 = R_1 + \gamma G_1$$

$$G_1 = R_2 + \gamma G_2$$

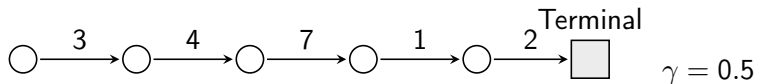
$$G_2 = R_3 + \gamma G_3$$

$$G_3 = R_4 + \gamma G_4$$

$$G_4 = R_5 + \gamma G_5$$

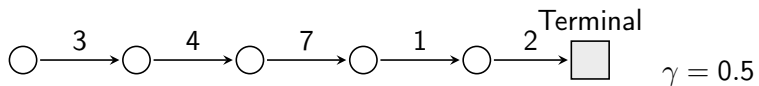
$$G_5 = 0$$

Computing returns efficiently



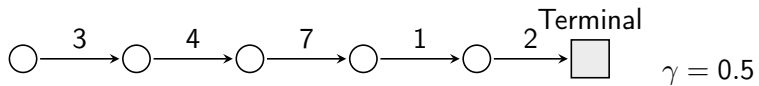
$$\begin{array}{rcl} G_4 & = & R_5 = 2 \\ & & G_5 = 0 \end{array}$$

Computing returns efficiently



$$G_3 = R_4 + \overbrace{\gamma G_4}^{\gamma R_5}$$
$$G_4 = R_5 = 2$$
$$G_5 = 0$$

Computing returns efficiently

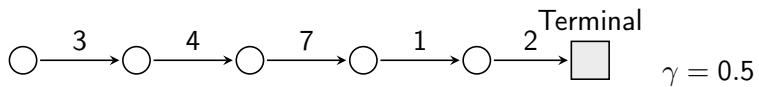


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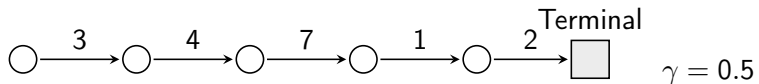
$$G_5 = 0$$

Computing returns efficiently



$$\begin{aligned} G_2 &= R_3 + \overbrace{\gamma R_4 + \gamma^2 R_5}^{\gamma G_3} \\ G_3 &= R_4 + \gamma R_5 = 2 \\ G_4 &= R_5 = 2 \\ G_5 &= 0 \end{aligned}$$

Computing returns efficiently



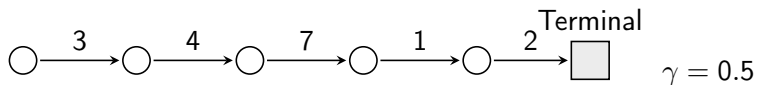
$$G_2 = R_3 + \gamma R_4 + \gamma^2 R_5 = 8$$

$$G_3 = R_4 + \gamma R_5 = 2$$

$$G_4 = R_5 = 2$$

$$G_5 = 0$$

Computing returns efficiently



$$G_0 = R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5 = 7$$

$$G_1 = R_2 + \gamma R_3 + \gamma^2 R_4 + \gamma^3 R_5 = 8$$

$$G_2 = R_3 + \gamma R_4 + \gamma^2 R_5 = 8$$

$$G_3 = R_4 + \gamma R_5 = 2$$

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6:      $G \leftarrow 0$ 
7:     for each step of episode,  $t = T - 1, T - 2, \dots, 0$  do
8:        $G \leftarrow \gamma G + R_{t+1}$ 
9:       if  $S_t \notin S_0, S_1, \dots, S_{t-1}$  then
10:         Append  $G$  to  $Ret(S_t)$ 
11:          $V(S_t) \leftarrow \text{average}(Ret(S_t))$ 
```

Monte Carlo Prediction Algorithms

Every-visit MC prediction

Require: a policy π to be evaluated

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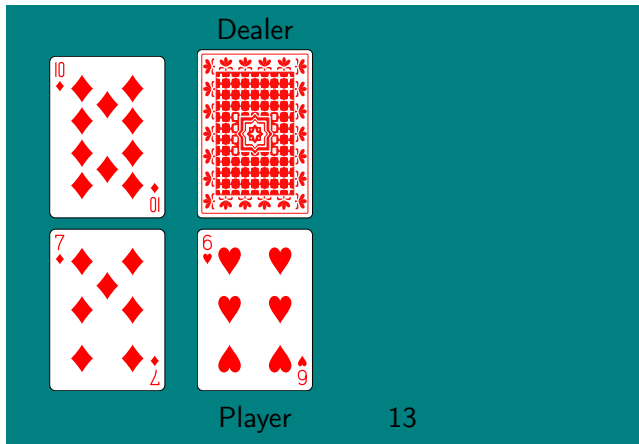
Blackjack example

- States (200 of them)
 - Current sum (12-21)
 - Dealer's showing card (ace-10)
 - Do I have a "useable" ace? (yes-no)
- Action **stick**: Stop receiving cards (and terminate)
- Action **hit**: Take another card (with replacement)
- Reward for **stick**:
 - +1 if sum of cards $>$ sum of dealer cards
 - 0 if sum of cards = sum of dealer cards
 - -1 if sum of cards $<$ sum of dealer cards
- Reward for **hit**:
 - -1 if sum of cards $>$ 21 (and terminate)
 - 0 otherwise
- Transitions: automatically **hit** if sum of cards $<$ 12
- Policy: **stick** when player's sum is 20 or 21, else **hit**



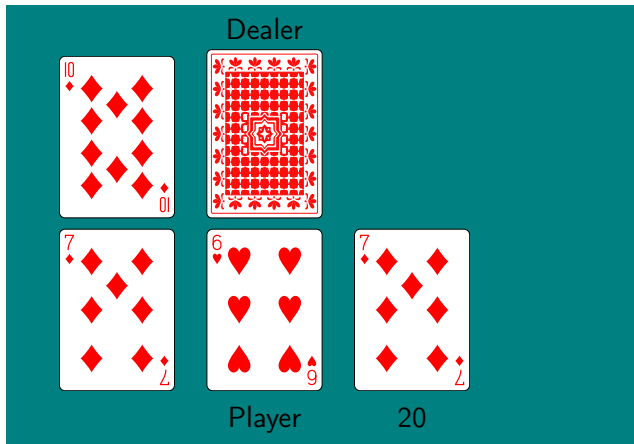
Blackjack Example

S (Ace, Sum, Dealer)	$Ret(S)$	$V(S)$



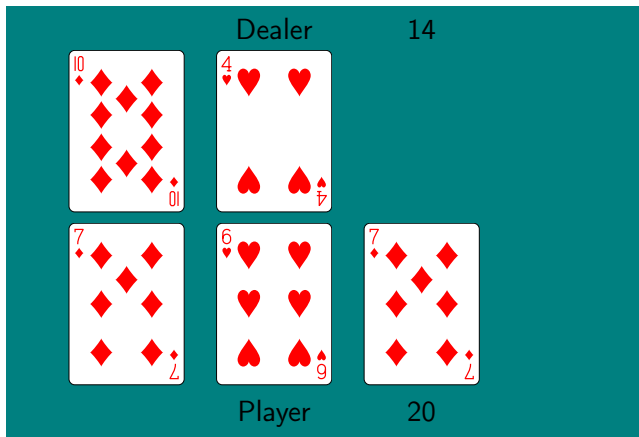
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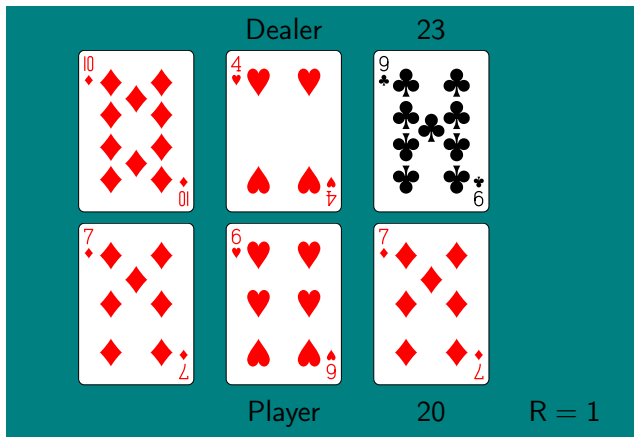
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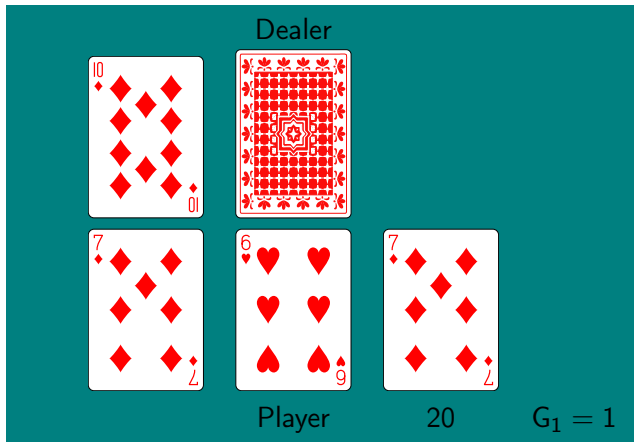
S (Ace, Sum, Dealer)	$Ret(S)$	$V(S)$



Blackjack Example

S (Ace, Sum, Dealer)	$Ret(S)$	$V(S)$
A	$Ret(A) = [1]$	1

$A = (\text{NoAce}, 20, 10)$

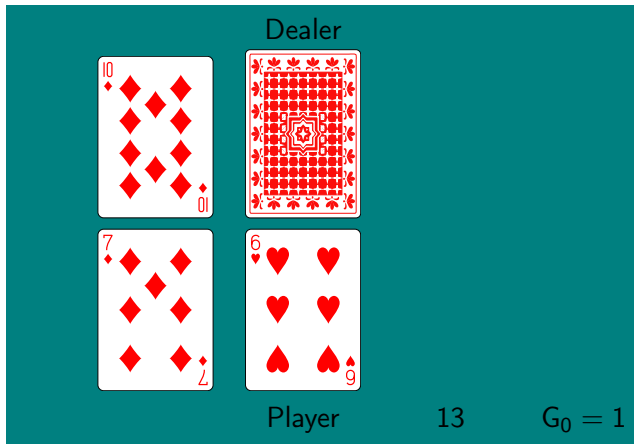


Blackjack Example

S (Ace, Sum, Dealer)	$Ret(S)$	$V(S)$
A	$Ret(A) = [1]$	1
B	$Ret(B) = [1]$	1

$A = (\text{NoAce}, 20, 10)$

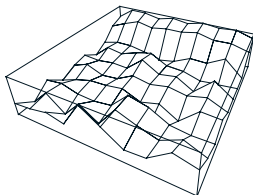
$B = (\text{NoAce}, 13, 10)$



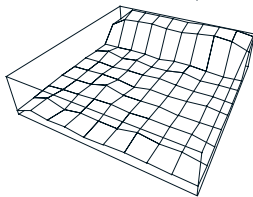
Blackjack value function

After 10,000 episodes

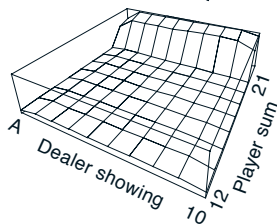
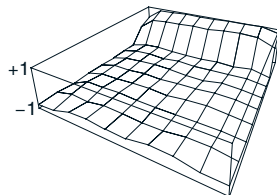
Usable
ace



No
usable
ace



After 500,000 episodes



Policy: **stick** if sum of cards ≥ 20 , otherwise **hit**

Incremental Mean

The mean $\mu_1, \mu_2, \dots$ of a sequence $x_1, x_2, \dots$ can be computed incrementally,

$$\begin{aligned}\mu_k &= \frac{1}{k} \sum_{j=1}^k x_j \\ &= \frac{1}{k} \left(x_k + \sum_{j=1}^{k-1} x_j \right) \\ &= \frac{1}{k} (x_k + (k-1) * \mu_{k-1}) \\ &= \mu_{k-1} + \frac{1}{k} (x_k - \mu_{k-1})\end{aligned}$$

Incremental Monte-Carlo Updates

- Update $V(s)$ incrementally after episode $S_1, A_1, R_2, \dots S_T$
- For each state S_t with return G_t

$$N(S_t) \leftarrow N(S_t) + 1$$

$$V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)}(G_t - V(S_t))$$

- In non-stationary problems, it can be useful to track a running mean, i.e. forget old episodes.

$$V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$$

Implications of Monte Carlo learning

- We do not need to keep a **large model** of the environment
- We estimate the value of an individual state **independently** of the values of other states (no bootstrapping)
- The **computation** needed to update the value of each state is independent on the size of the MDP (state-space/transition dynamics)

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Temporal-Difference Learning

- TD methods learn directly from episodes of experience
- TD is **model-free**: no knowledge of MDP transitions / rewards
- TD learns from **incomplete** episodes, by **bootstrapping**
- TD updates a guess towards a guess

Review: Estimating Values from Returns

$$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t \mid S_t = s]$$

Review: Estimating Values from Returns

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Review: Estimating Values from Returns

$$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t \mid S_t = s]$$

$$V(S_t) \leftarrow V(S_t) + \alpha \left(\underbrace{G_t}_{1\text{-step?}} - V(S_t) \right)$$

Bootstrapping

$$G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

Bootstrapping

$$\begin{aligned} G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

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$$\begin{aligned} G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

$$v_\pi(s) \doteq \mathbb{E}_\pi [G_t \mid S_t = s]$$

Bootstrapping

$$\begin{aligned} G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

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Bootstrapping

$$\begin{aligned} G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

$$\begin{aligned} v_\pi(s) &\doteq \mathbb{E}_\pi [G_t \mid S_t = s] \\ &= \mathbb{E} [R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= R_{t+1} + \gamma v_\pi(S_{t+1}) \end{aligned}$$

- Goal: learn v_π online from experience under policy π
- Incremental every-visit Monte-Carlo
 - Update value $V(S_t)$ toward **actual** return G_t

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

- Simplest temporal-difference learning algorithm: TD(0)
 - Update value $V(S_t)$ toward **estimated** return $R_{t+1} + \gamma V(S_{t+1})$

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

- $R_{t+1} + \gamma V(S_{t+1})$ is called the **TD target**
- $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ is called the **TD error**

Temporal Difference

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

Recall

$$G_t \approx R_{t+1} + \gamma V(S_{t+1})$$

TD Policy Evaluation

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

TD Policy Evaluation

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

Require: π a policy to be evaluated

```
1: function policyEvaluation( $\pi$ )
2:    $V \leftarrow V' \leftarrow$  initialize to zeroes
3:   repeat
4:      $\Delta \leftarrow 0$ 
5:     for each  $s \in \mathcal{S}$  do
6:        $V'(s) \leftarrow \sum_{a \in \mathcal{A}} \pi(a | s) \sum_{s', r} \mathcal{P}(s', r | s, a) [r + \gamma V(s')]$ 
7:        $\Delta \leftarrow \max(\Delta, |V'(s) - v(s)|)$ 
8:      $V \leftarrow V'$ 
9:   until  $\Delta < \theta$ 
10:  return  $V \approx v_\pi$ 
```

▷ A small positive number

TD Policy Evaluation

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

Require: π a policy to be evaluated

```
1: function policyEvaluation( $\pi$ )
2:    $V \leftarrow V' \leftarrow$  initialize to zeroes
3:   repeat
4:      $\Delta \leftarrow 0$ 
5:     for each  $s \in \mathcal{S}$  do
6:        $V'(s) \leftarrow \sum_{a \in \mathcal{A}} \pi(a | s) \sum_{s', r} \mathcal{P}(s', r | s, a) [r + \gamma V(s')]$ 
7:        $\Delta \leftarrow \max(\Delta, |V'(s) - v(s)|)$ 
8:      $V \leftarrow V'$ 
9:   until  $\Delta < \theta$ 
10:  return  $V \approx v_\pi$ 
```

▷ A small positive number

Tabular TD

Require: π a policy to be evaluated, $\alpha \in (0, 1]$ step size

```
1: function tdPolicyEvaluation( $\pi, \alpha$ )  
2:    $V(s)$  initialized arbitrarily  $\forall s \in \mathcal{S}$  except terminals  
3:   loop for each episode  
4:     initialize  $S$   
5:     repeat  
6:        $A \leftarrow \pi(S)$   
7:       Take action  $A$ , observe  $R, S'$   
8:        $V(S) \leftarrow V(S) + \alpha [R + \gamma V(S') - V(S)]$   
9:        $S \leftarrow S'$   
10:    until  $S$  is terminal
```

Driving Home Example (MC)



Driving Home Example (MC)

leave



0 mins
elapsed



Driving Home Example (MC)

leave



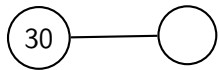
0 mins
elapsed



Driving Home Example (MC)

leave

exit



0 mins
elapsed

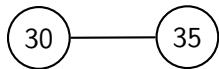
5 mins
elapsed



Driving Home Example (MC)

leave

exit



0 mins
elapsed

5 mins
elapsed

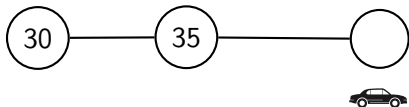


Driving Home Example (MC)

leave

exit

exit highway



0 mins
elapsed

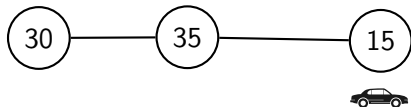
5 mins
elapsed

20 mins
elapsed



Driving Home Example (MC)

leave exit exit highway



0 mins
elapsed

5 mins
elapsed

20 mins
elapsed



Driving Home Example (MC)

leave

exit

exit highway

secondary road

30

35

15



0 mins
elapsed

5 mins
elapsed

20 mins
elapsed

30 mins
elapsed

Driving Home Example (MC)

leave

exit

exit highway

secondary road

30

35

15

10



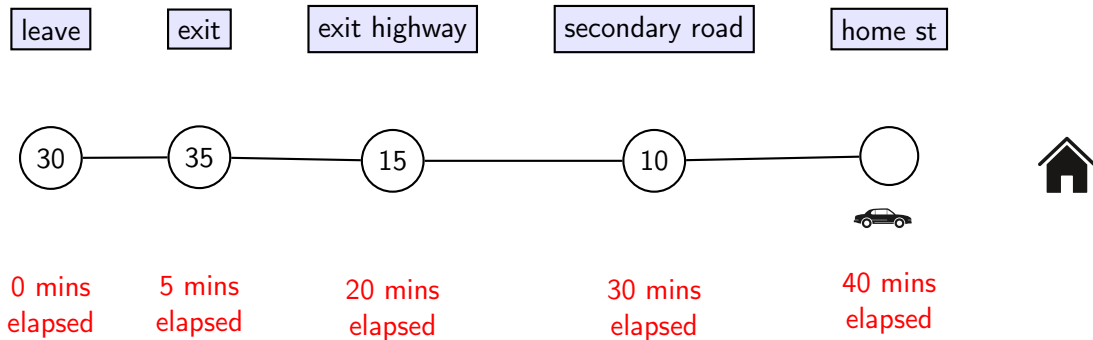
0 mins
elapsed

5 mins
elapsed

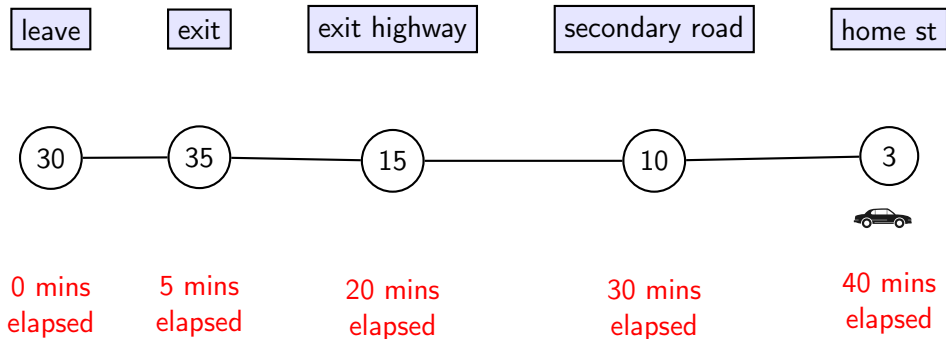
20 mins
elapsed

30 mins
elapsed

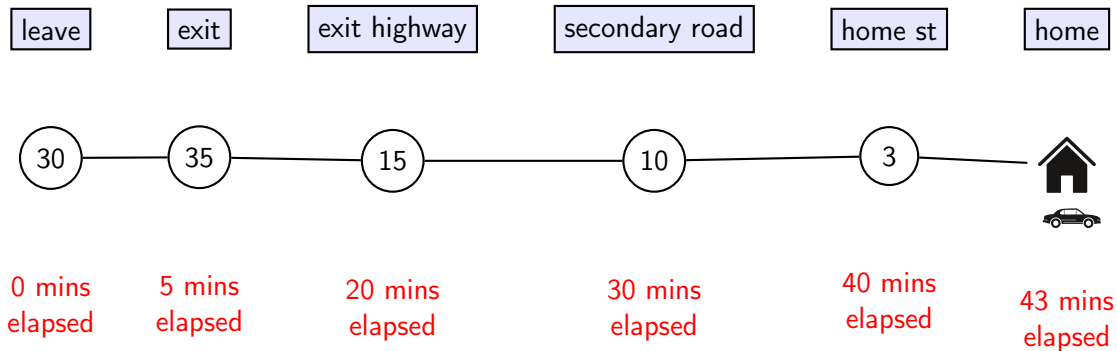
Driving Home Example (MC)



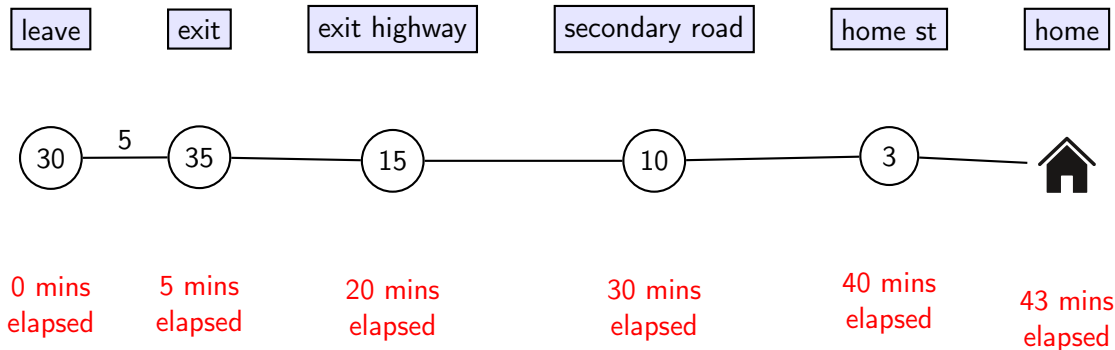
Driving Home Example (MC)



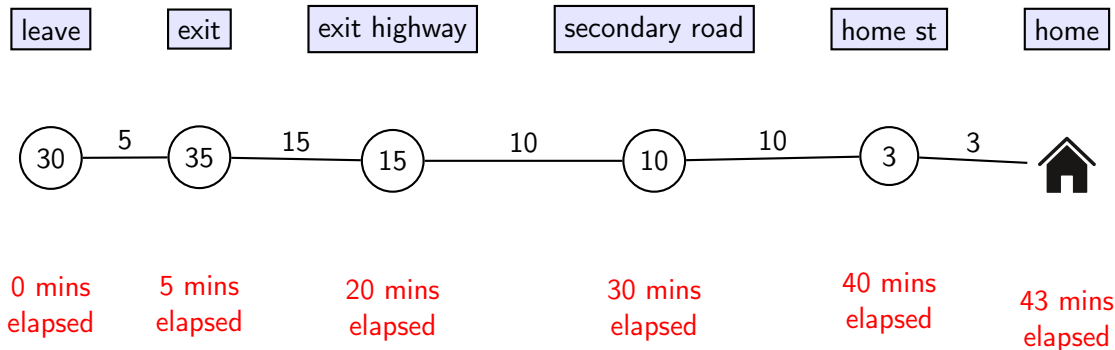
Driving Home Example (MC)



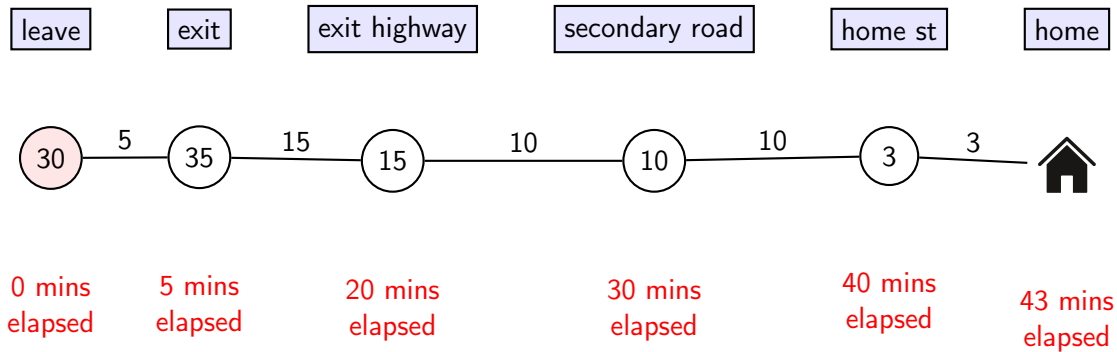
Driving Home Example (MC)



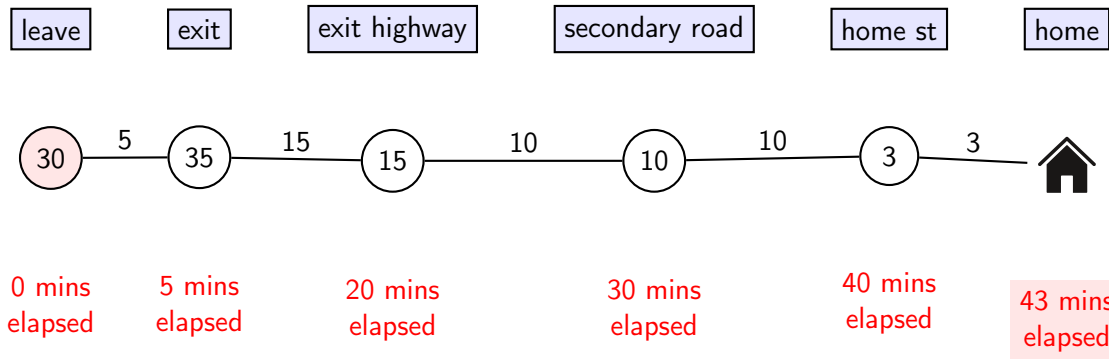
Driving Home Example (MC)



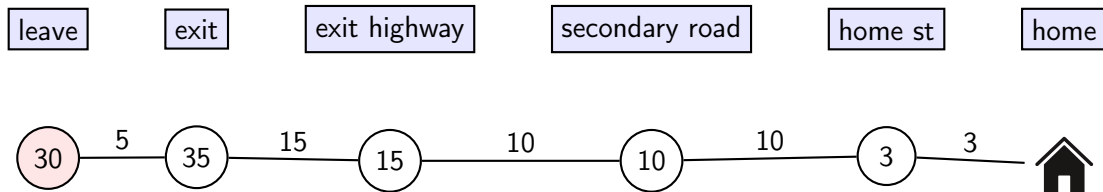
Driving Home Example (MC)



Driving Home Example (MC)

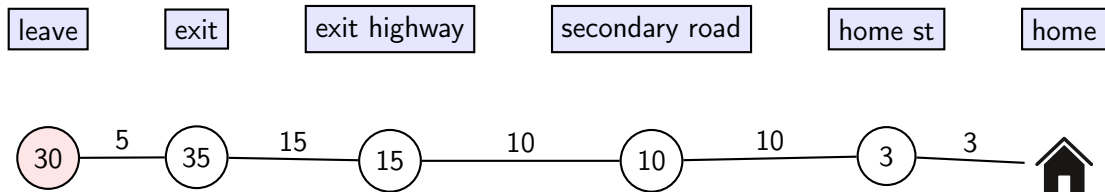


Driving Home Example (MC)



$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

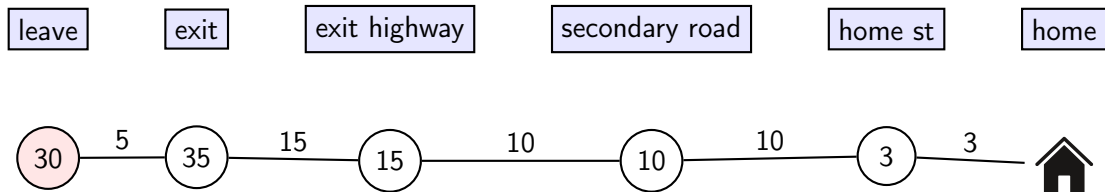
Driving Home Example (MC)



$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$\alpha = 1 \quad \gamma = 1$$

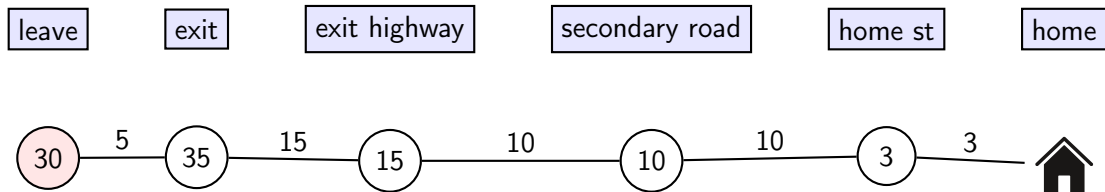
Driving Home Example (MC)



$$V(S_t) \leftarrow V(S_t) + \alpha (\textcolor{red}{G}_t - V(S_t))$$

$$\alpha = 1 \quad \gamma = 1$$

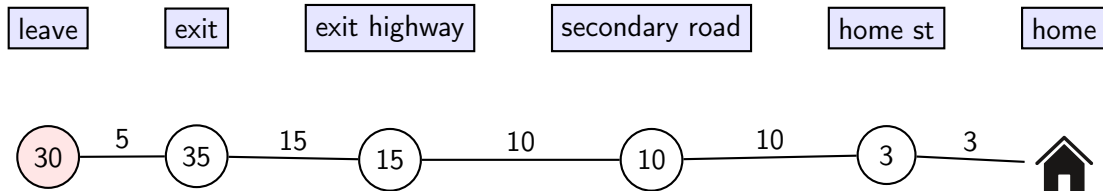
Driving Home Example (MC)



$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$
$$V(\text{leave}) \leftarrow V(\text{leave}) + \alpha (G_0 - V(\text{leave}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_0 = 5 + 15 + 10 + 10 + 3 = 43$$

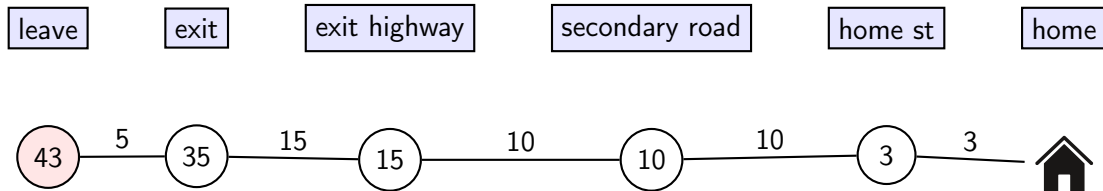
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{leave}) \leftarrow V(\text{leave}) + \alpha (G_0 - V(\text{leave}))$$

$$\leftarrow 30 + 43 - 30$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_0 = 5 + 15 + 10 + 10 + 3 = 43$$

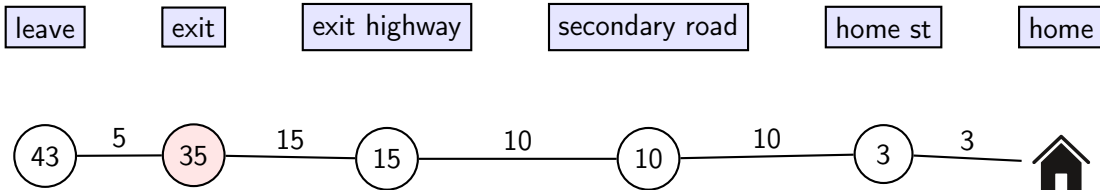
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{leave}) \leftarrow V(\text{leave}) + \alpha (G_0 - V(\text{leave}))$$

$$\leftarrow 30 + 43 - 30$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_1 = 15 + 10 + 10 + 3 = 38$$

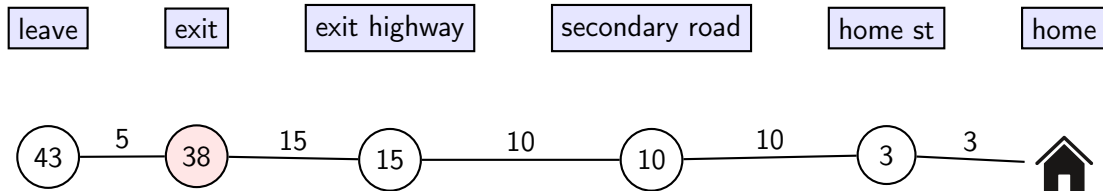
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{exit}) \leftarrow V(\text{exit}) + \alpha (G_1 - V(\text{exit}))$$

$$\leftarrow 35 + 38 - 35$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_1 = 15 + 10 + 10 + 3 = 38$$

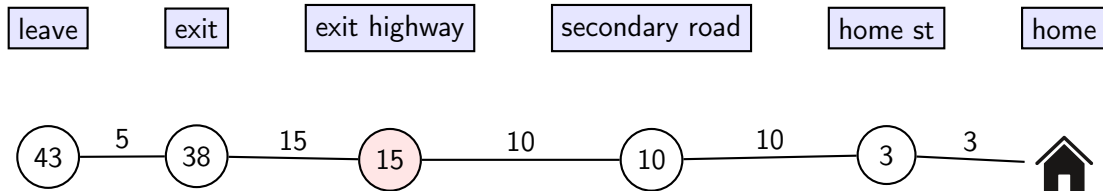
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{exit}) \leftarrow V(\text{exit}) + \alpha (G_1 - V(\text{exit}))$$

$$\leftarrow 35 + 38 - 35$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_2 = 10 + 10 + 3 = 23$$

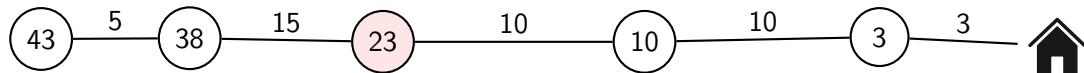
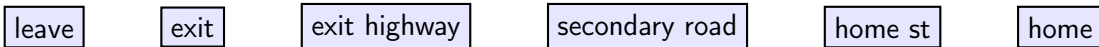
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{exit highway}) \leftarrow V(\text{exit highway}) + \alpha (G_2 - V(\text{exit highway}))$$

$$\leftarrow 15 + 23 - 15$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_2 = 10 + 10 + 3 = 23$$

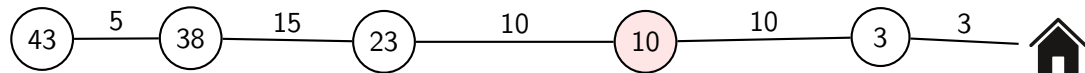
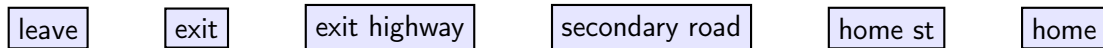
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{exit highway}) \leftarrow V(\text{exit highway}) + \alpha (G_2 - V(\text{exit highway}))$$

$$\leftarrow 15 + 23 - 15$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_3 = 10 + 3 = 13$$

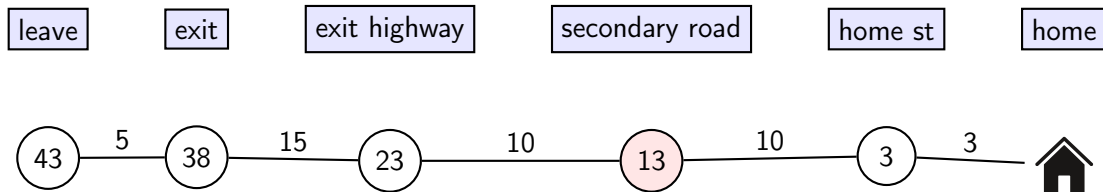
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{secondary road}) \leftarrow V(\text{secondary road}) + \alpha (G_3 - V(\text{secondary road}))$$

$$\leftarrow 10 + 13 - 10$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$G_3 = 10 + 3 = 13$$

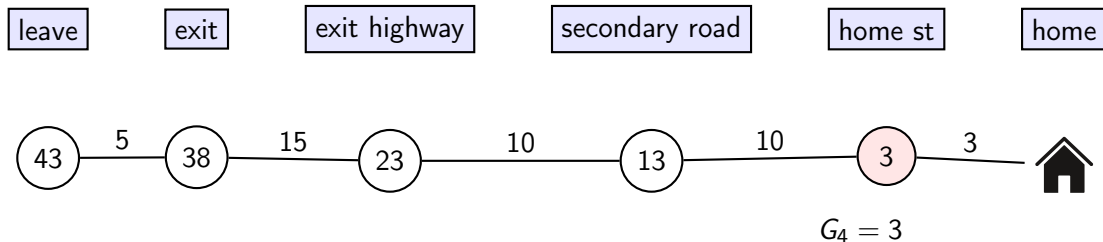
$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$V(\text{secondary road}) \leftarrow V(\text{secondary road}) + \alpha (G_3 - V(\text{secondary road}))$$

$$\leftarrow 10 + 13 - 10$$

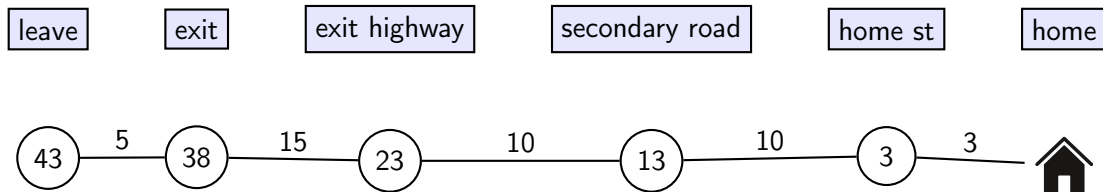
$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (MC)



$$\begin{aligned} V(S_t) &\leftarrow V(S_t) + \alpha (G_t - V(S_t)) \\ V(\text{home}) &\leftarrow V(\text{home}) + \alpha (G_4 - V(\text{home})) \\ &\leftarrow 3 + 3 - 3 \\ \alpha &= 1 \quad \gamma = 1 \end{aligned}$$

Driving Home Example (MC)



$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$\alpha = 1 \quad \gamma = 1$$

Must wait until the end of the episode before
learning can begin!

Driving Home Example (TD)

leave

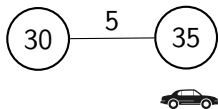


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave



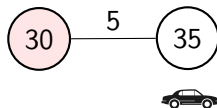
$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave

exit



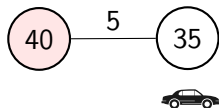
$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$
$$V(\text{leave}) \leftarrow V(\text{leave}) + \alpha (R_1 + \gamma V(\text{exit}) - V(\text{leave}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave

exit



$$\begin{aligned} V(S_t) &\leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t)) \\ V(\text{leave}) &\leftarrow V(\text{leave}) + \alpha (R_1 + \gamma V(\text{exit}) - V(\text{leave})) \\ &\leftarrow 30 + (5 + 35 - 30) \end{aligned}$$

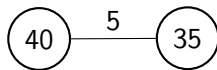
$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave

exit

exit highway



$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$
$$V(\text{exit}) \leftarrow V(\text{exit}) + \alpha (R_2 + \gamma V(\text{exit highway}) - V(\text{exit}))$$

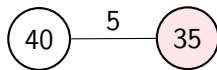
$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave

exit

exit highway



$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$
$$V(\text{exit}) \leftarrow V(\text{exit}) + \alpha (R_2 + \gamma V(\text{exit highway}) - V(\text{exit}))$$

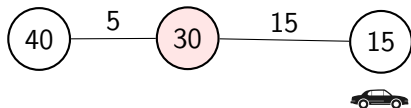
$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

leave

exit

exit highway



$$\begin{aligned} V(S_t) &\leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t)) \\ V(\text{exit}) &\leftarrow V(\text{exit}) + \alpha (R_2 + \gamma V(\text{exit highway}) - V(\text{exit})) \\ &\leftarrow 35 + (15 + 15 - 35) \end{aligned}$$

$$\alpha = 1 \quad \gamma = 1$$

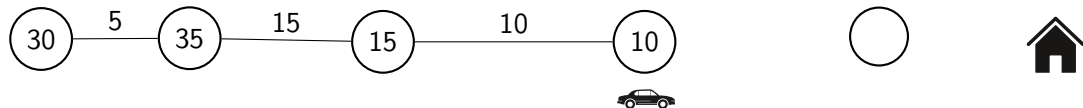
Driving Home Example (TD)

leave

exit

exit highway

secondary road



$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{exit highway}) \leftarrow V(\text{exit highway}) + \alpha (R_3 + \gamma V(\text{secondary road}) - V(\text{exit highway}))$$

$$\alpha = 1 \quad \gamma = 1$$

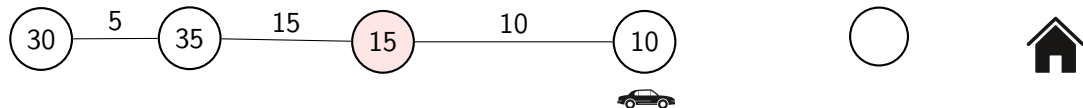
Driving Home Example (TD)

leave

exit

exit highway

secondary road



$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{exit highway}) \leftarrow V(\text{exit highway}) + \alpha (R_3 + \gamma V(\text{secondary road}) - V(\text{exit highway}))$$

$$\alpha = 1 \quad \gamma = 1$$

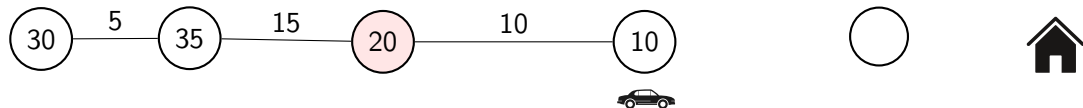
Driving Home Example (TD)

leave

exit

exit highway

secondary road

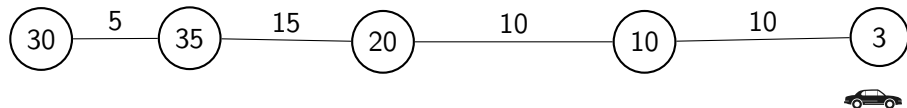
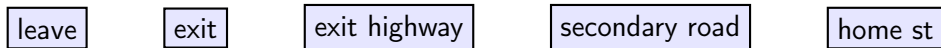


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$\begin{aligned} V(\text{exit highway}) &\leftarrow V(\text{exit highway}) + \alpha (R_3 + \gamma V(\text{secondary road}) - V(\text{exit highway})) \\ &\leftarrow 15 + (10 + 10 - 15) \end{aligned}$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

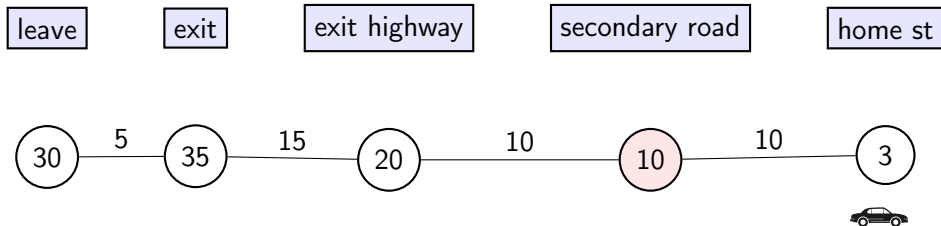


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{secondary road}) \leftarrow V(\text{secondary road}) + \alpha (R_4 + \gamma V(\text{home st}) - V(\text{secondary road}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

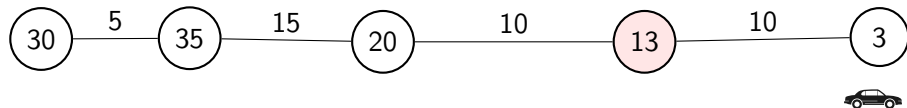
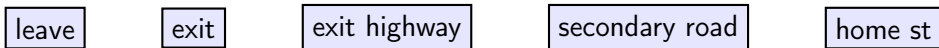


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{secondary road}) \leftarrow V(\text{secondary road}) + \alpha (R_4 + \gamma V(\text{home st}) - V(\text{secondary road}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

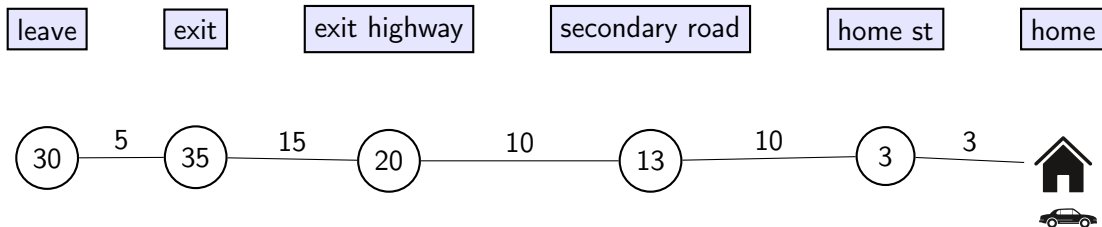


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$\begin{aligned} V(\text{secondary road}) &\leftarrow V(\text{secondary road}) + \alpha (R_4 + \gamma V(\text{home st}) - V(\text{secondary road})) \\ &\leftarrow 10 + (10 + 3 - 10) \end{aligned}$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

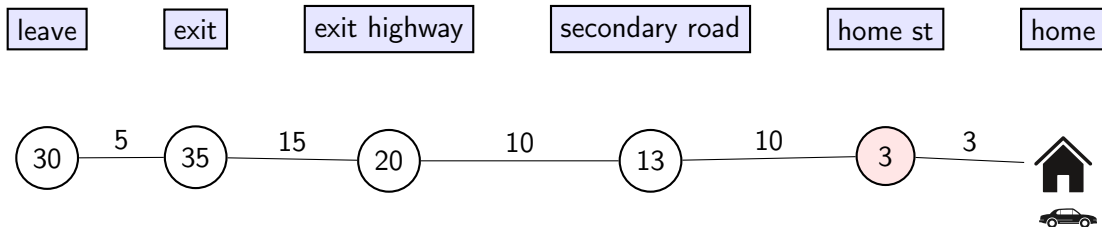


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{home st}) \leftarrow V(\text{home st}) + \alpha (R_5 + \gamma V(\text{home}) - V(\text{home st}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

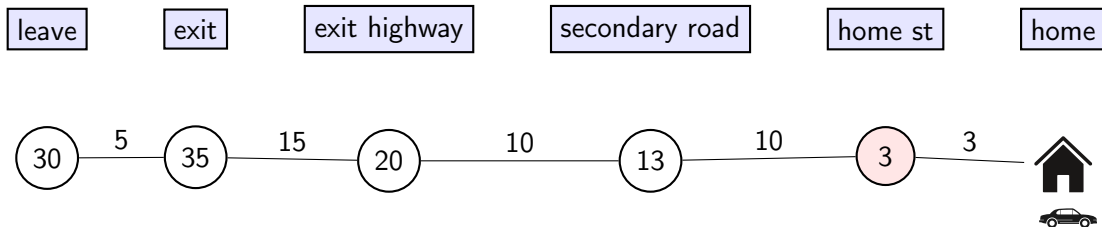


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$V(\text{home st}) \leftarrow V(\text{home st}) + \alpha (R_5 + \gamma V(\text{home}) - V(\text{home st}))$$

$$\alpha = 1 \quad \gamma = 1$$

Driving Home Example (TD)

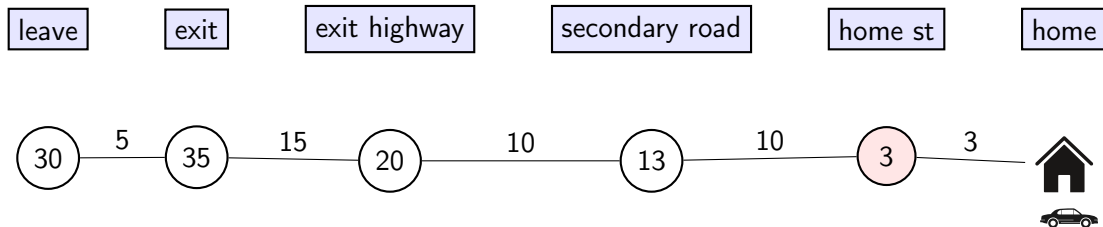


$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

$$\begin{aligned} V(\text{home st}) &\leftarrow V(\text{home st}) + \alpha (R_5 + \gamma V(\text{home}) - V(\text{home st})) \\ &\leftarrow 3 + (3 + 0 - 3) \end{aligned}$$

$$\alpha = 1 \quad \gamma = 1$$

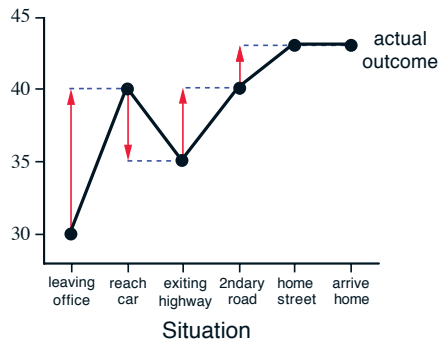
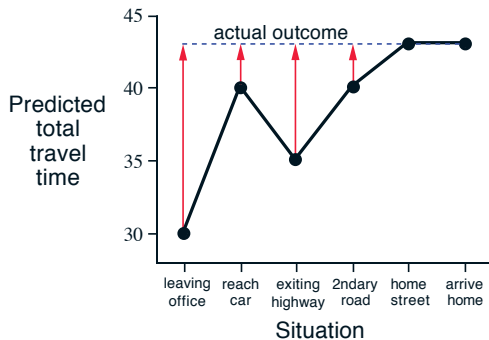
Driving Home Example (TD)



$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

We can learn online without waiting
for the episode to end!

Driving Home Example: MC vs. TD



Outline

- 1 Recap
- 2 Monte-Carlo Learning
- 3 Temporal-Difference Learning
- 4 Comparison

Advantages and Disadvantages of MC vs. TD

- TD can learn before knowing the final outcome
 - TD can learn online after every step
 - MC must wait until end of episode before we know the return
- TD can learn without the final outcome TD can learn from incomplete sequences
 - MC can only learn from complete sequences
 - TD works in continuing (non-terminating) environments
 - MC only works for episodic (terminating) environments

Bias/Variance Tradeoff

- Return $G_t = R_{t+1} + \gamma R_{t+2} + \dots \gamma^{T-1} R_T$ is **unbiased** estimate of $v_\pi(S_t)$
- True TD target $R_{t+1} + \gamma v_\pi(S_{t+1})$ is **unbiased** estimate of $v_\pi(S_t)$
- Actual TD target $R_{t+1} + \gamma V(S_{t+1})$ is **biased** estimate of $v_\pi(S_t)$
- TD target is much lower variance than return:
 - Return depends on **many** random actions, transitions, rewards
 - TD target depends on **one** random action, transition, reward

Advantages and Disadvantages of MC vs. TD (2)

- MC has high variance, zero bias
 - Good convergence properties (even with function approximation)
 - Not very sensitive to initial value
 - Very simple to understand and use
- TD has low variance, some bias
 - Usually more efficient than MC (in practice)
 - TD(0) converges to $v_{\pi}(s)$ (but not always with function approximation)
 - More sensitive to initial value

Random Walk Example



MRP with uniform probability for left and right movement.

Random Walk Example



MRP with uniform probability for left and right movement.

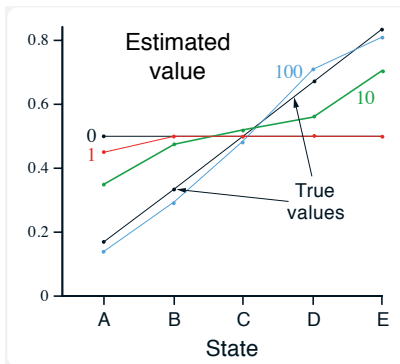
True values: $v_{\pi}(A) = \frac{1}{6}$, $v_{\pi}(B) = \frac{2}{6}$, $v_{\pi}(C) = \frac{3}{6}$, $v_{\pi}(D) = \frac{4}{6}$, $v_{\pi}(E) = \frac{5}{6}$

Random Walk Example



MRP with uniform probability for left and right movement.

True values: $v_{\pi}(A) = \frac{1}{6}$, $v_{\pi}(B) = \frac{2}{6}$, $v_{\pi}(C) = \frac{3}{6}$, $v_{\pi}(D) = \frac{4}{6}$, $v_{\pi}(E) = \frac{5}{6}$

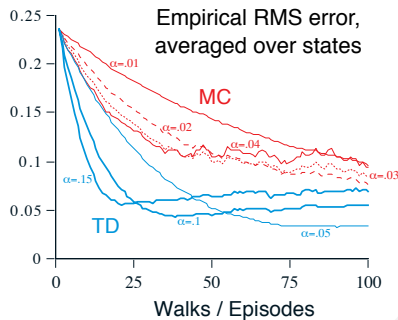
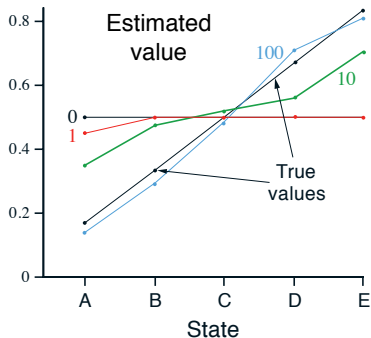


Random Walk Example



MRP with uniform probability for left and right movement.

True values: $v_{\pi}(A) = \frac{1}{6}$, $v_{\pi}(B) = \frac{2}{6}$, $v_{\pi}(C) = \frac{3}{6}$, $v_{pi}(D) = \frac{4}{6}$, $v_{\pi}(E) = \frac{5}{6}$



Batch MC and TD

- MC and TD converge: $V(s) \rightarrow v_\pi(s)$ as experience $\rightarrow \infty$
- But what about batch solution for finite experience?

$$\begin{array}{c} s_1^1, a_1^1, r_2^1, \dots, s_{T_1}^1 \\ \vdots \\ s_1^K, a_1^K, r_2^K, \dots, s_{T_K}^K \end{array}$$

- e.g. Repeatedly sample episode $k \in [1, K]$
- Apply MC or TD(0) to episode k

AB Example

Two states A,B; no discounting; 8 episodes of experience

A, 0, B, 0

B, 1

B, 1

B, 1

B, 1

B, 1

B, 1

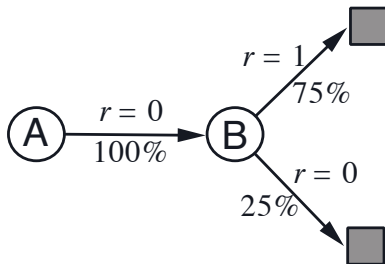
B, 0

What is $V(A)$, $V(B)$?

AB Example

Two states A,B; no discounting; 8 episodes of experience

A, 0, B, 0
B, 1
B, 1
B, 1
B, 1
B, 1
B, 1
B, 0



What is $V(A)$, $V(B)$?

We can all agree that $V(B) = \frac{3}{4}$

Certainty Equivalence

- MC converges to solution with minimum mean-squared error
 - Best fit to the observed returns

$$\sum_{k=1}^K \sum_{t=1}^{T_k} \left(G_r^k - V(s_t^k) \right)^2$$

- In the AB example, $V(A) = 0$
- TD(0) converges to solution of max likelihood Markov model
 - Solution to the MDP $\langle \mathcal{S}, \mathcal{A}, \hat{\mathcal{P}}, \hat{\mathcal{R}}, \gamma \rangle$ that best fits the data

$$\hat{\mathcal{P}}_{s,s'}^a = \sum_{k=1}^K \sum_{t=1}^{T_k} \frac{1}{N(s,a)} \mathbf{1}(s_t^k, a_t^k, s_{t+1}^k = s, a, s')$$

$$\hat{\mathcal{R}}_s^a = \sum_{k=1}^K \sum_{t=1}^{T_k} \frac{1}{N(s,a)} \mathbf{1}(s_t^k, a_t^k = s, a) r_t^k$$

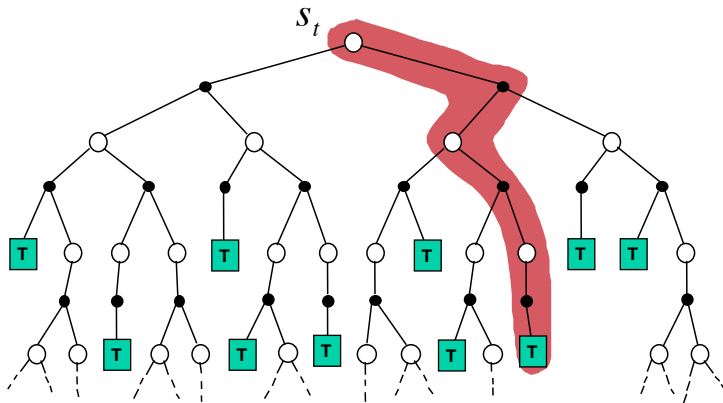
- In the AB example, $V(A) = 0.75$

Advantages and Disadvantages of MC vs. TD (3)

- TD exploits Markov property
 - Usually more efficient in Markov environments
- MC does not exploit Markov property
 - Usually more effective in non-Markov environments

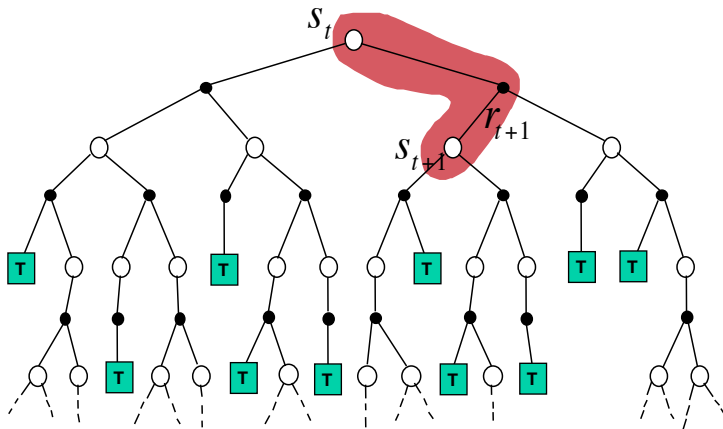
Unified View: Monte-Carlo Backup

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$



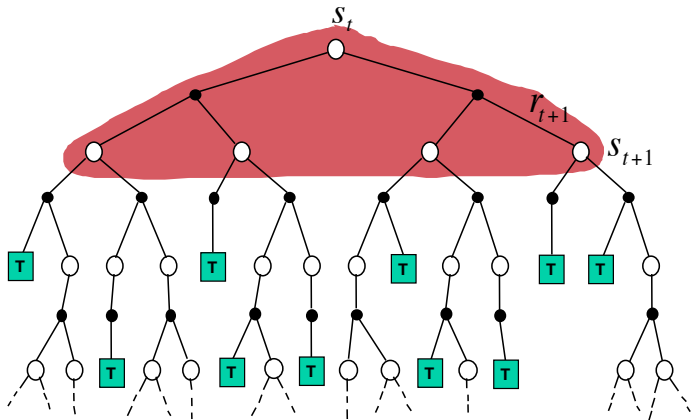
Unified View: Temporal-Difference Backup

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} - \gamma V(S_{t+1}) - V(S_t))$$



Unified View: Dynamic Programming Backup

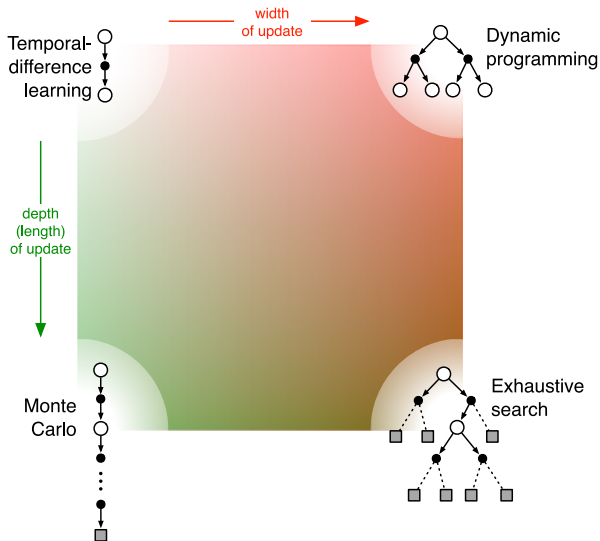
$$V(S_t) \leftarrow \mathbb{E} [R_{t+1} + \gamma V(S_{t+1})]$$



Bootstrapping and Sampling

- **Bootstrapping:** update involves an estimate
 - MC does not bootstrap
 - DP bootstraps
 - TD bootstraps
- **Sampling:** update samples an expectation
 - MC samples
 - DP does not sample
 - TD samples

Unified View of Reinforcement Learning



Conclusions

- Key differences:
 - Monte-Carlo computes estimates based on true-values
 - Temporal-Difference computes estimates based on other estimates
- Both MC and TD converge to the true values asymptotically
 - Temporal-Difference tends to converge faster
 - Monte-Carlo is easier to implement for non-tabular RL
- We can interpolate between these two techniques using n-step bootstrapping (Lecture 7)

Lecture Summary

- Prediction methods in Reinforcement Learning
 - Monte-Carlo Sampling
 - Temporal Difference
- Bootstrapping
- Comparison